

Five Must-Haves when Choosing a Software Development Partner



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The Application Economy

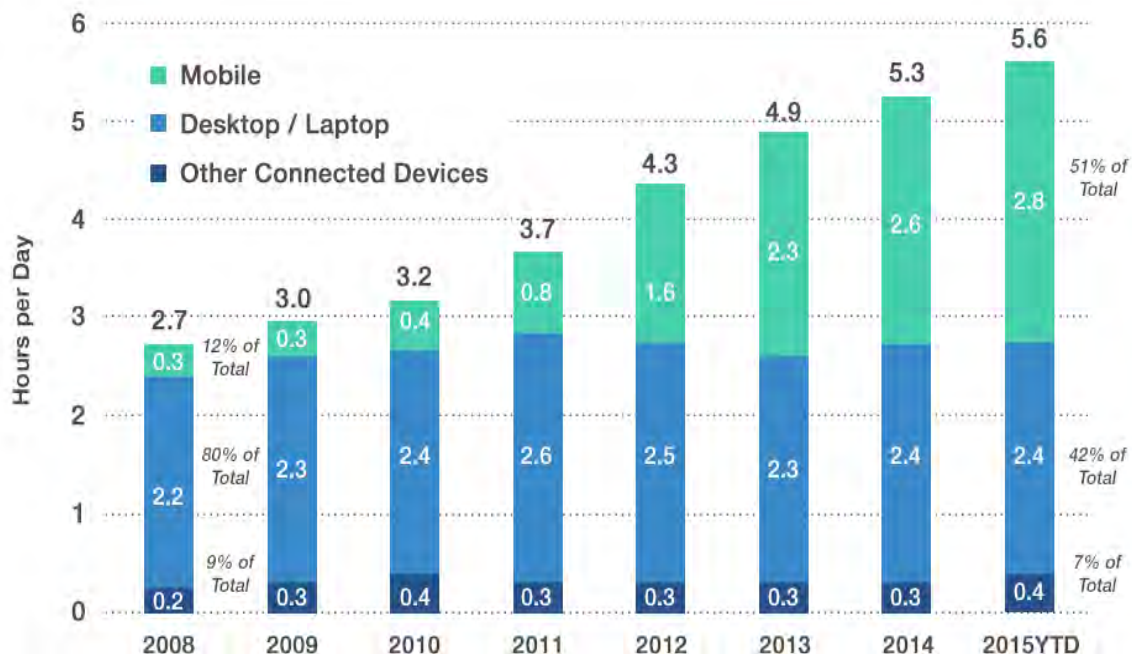
In our increasingly application-driven economy, delivering and deploying applications and services that business units demand has put IT and development teams under increasing pressure to deliver. To meet that challenge, many businesses are partnering with outside market-savvy software development teams to meet the bottom-line goals of a competitive market.

This whitepaper looks into the way the development process has changed, how the mega-trends of DevOps adoption, IT consumerization, and the Cloud have impacted the way we work, and what you should look for when making the decision to use a software development partner to meet business-critical demands.

“There’s an app for that” has gone from a catch phrase from the iTunes store to the very heart of business computing. The past few years has seen a seismic shift in the way users interact with corporate data thanks to the

increasing ubiquity of mobile devices from tablet to smartphone. What’s more, mobile devices are now more often being used to create – not just consume – business information, and the number of new devices used to interact with business data is poised to explode thanks to the internet of things (IoT). What will the impact be? Business Insider forecasts 34 billion ‘things’ connected by 2020, up from 10 billion in 2015.¹

The application explosion isn’t limited to new mobile devices. Business adoption of bring your own device (BYOD) is accelerating, enabling employees to utilize personally owned devices, whether Windows, Mac, iOS, or Android for business use. And business users, dazzled by the broad array of software-as-a-service (SaaS) offerings, expect IT departments to deliver custom applications for virtually every business process at the drop of a hat. The challenges that presents are myriad.



Source: KPCB and Mary Meeker's 2015 Internet Trends Study
<http://www.lsain Insider.com/study-us-adult-daily-time-spent-with-digital-media-grows-2-5-hours-in-5-years/archives>

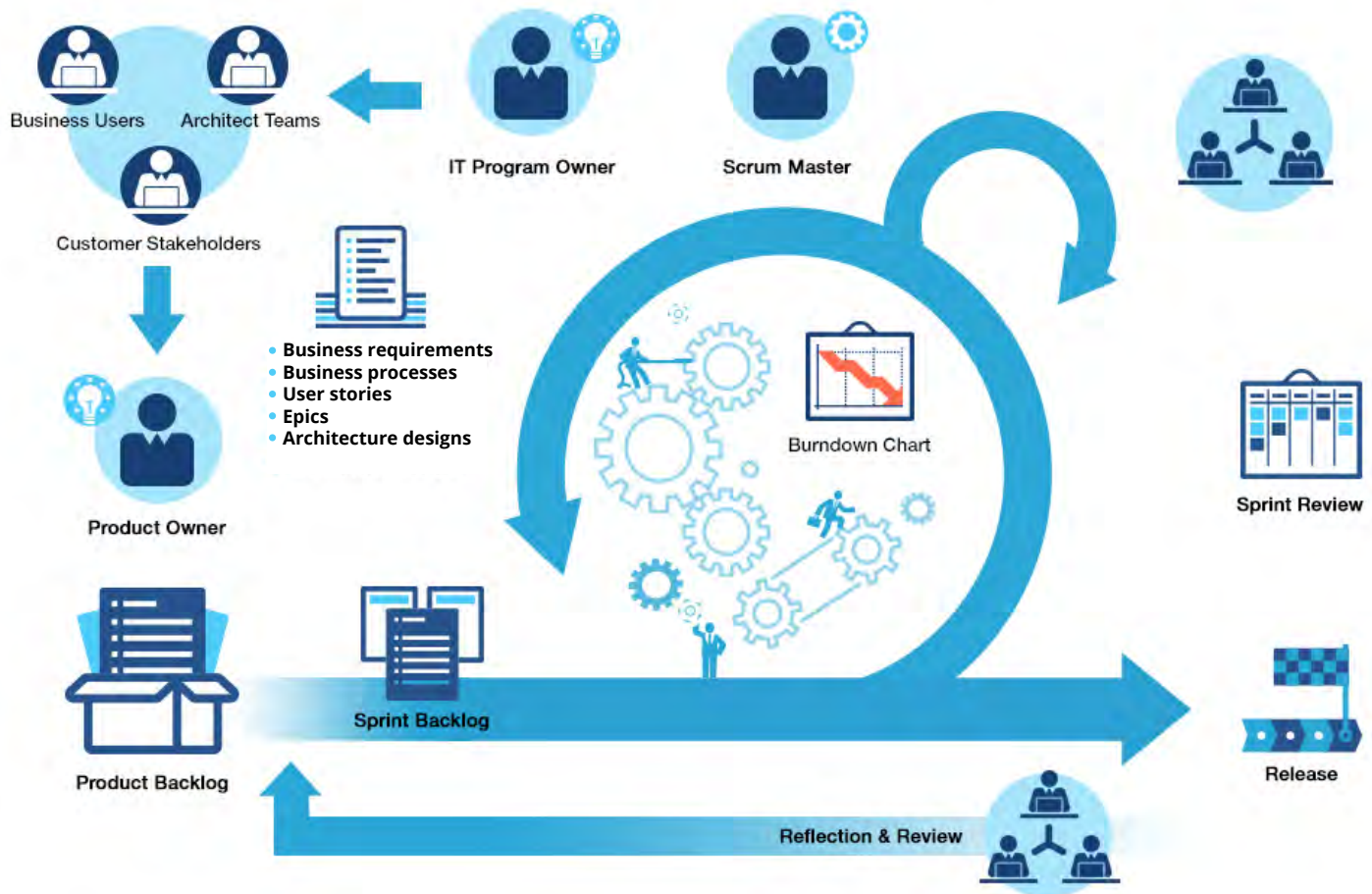
The Evolution of Development

Increasing demand for new applications has forced a shift in the way we develop code to keep up with the demand. Perhaps the biggest shift is from traditional 'waterfall' development methodologies to 'agile.' Instead of coding an entire app, testing, and releasing, agile teams work on small pieces of code functionality, frequently release code, and teams meet daily to gauge progress and reprioritize based on user and testing feedback.

Agile methodology and the Scrum process have had a major impact on the way software is developed. Instead of so-called waterfall methodology where major releases were infrequent, agile shops are in a 'constant state of release,' breaking development tasks down into small chunks or 'sprints.'

Although increasingly growing in popularity, agile is not the only way to develop code, many waterfall shops still thrive.

Agile Methodology - Scrum Process

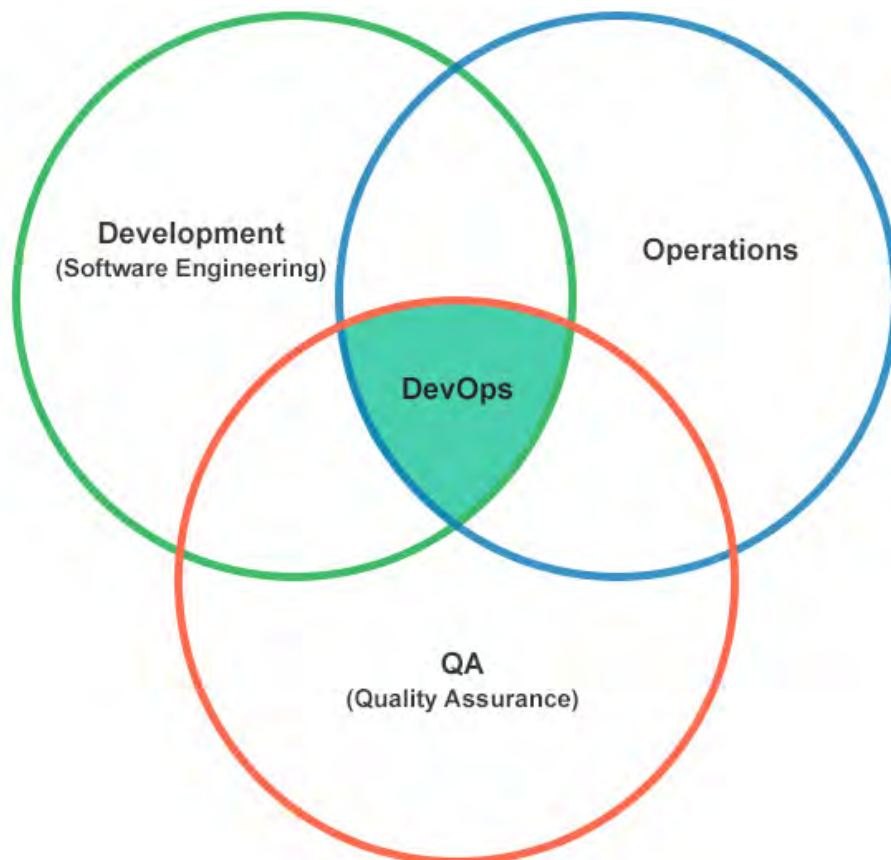


Acceleration of DevOps Adoption

Often when adopting an agile approach, organizations also move to a DevOps philosophy, a set of processes and methods for collaboration and communicating between developers, quality assurance/testing, and IT operations. The acceleration of DevOps adoption is also an outcome of the increasing volume of applications and platforms, but other factors are also in play, particularly the continuing move to cloud-based platforms for a wide range of applications and data. In an environment where applications and data can reside in multiple places, on multiple platforms, and be accessed by a broad variety of devices from anywhere

there is broadband, the need to integrate testing, dev teams, and operations becomes clear. There are just too many moving parts to maintain a business-as-usual stance.

DevOps has rapidly gained in adoption, as it promises to accelerate delivery of quality code while bridging the gaps between development, QA/testing and operations teams. Adopting DevOps practices has been a logical next step for many organizations who have moved to an agile development environment, although one is not a prerequisite for the other.



Consumerization of IT

Widespread mobile device adoption has led to another trend dramatically impacting the way we think about applications – the consumerization of IT. Business users expect the same kind of user interface and experience that they receive from consumer-focused apps such as banking or social networking. Smart businesses have evolved their out-of-box experience (OOBE) for software into a ‘design thinking’ strategy that focuses as much on how to reduce keystrokes, gestures and eye movements as it does on functionality and back-end data access.

Ideally, all business applications should be consistent in their look and feel, intuitive to use as users switch from device to device, and be creative and esthetically pleasing without looking like it was quickly cobbled from some template.



These seismic shifts, while driving user adoption and efficiency, come at a cost. The shift to mobile, agile and DevOps can bring challenges, as many organizations lack the resources or experience necessary to achieve this transformation. Often, when faced with an IT organization unable to meet these new business demands, line-of-business (LOB) users will turn to public cloud providers or SaaS vendors to gain the new functionality they desire. This can lead to the rise of “shadow IT” in an organization, effectively creating IT within the business unit as well.

“When you have a complex system it’s easy for things to go off kilter,” says Mike Sandler, a senior technology executive and veteran of numerous start-ups. “I’ve often relied on very talented off-shore resources to help get the job done.”

There are many reasons why organizations turn to a partner for custom software development projects. Rapidly changing environments often require new skills and tools that are unfamiliar to internal development teams. Just keeping the lights on can be hard enough; learning new languages, platforms and design approaches can put an onerous burden on any organization. Today, there are seemingly an infinite number of companies who want to be your development partner, so in the pages to follow we will present the five most critical factors to consider when choosing to partner and when choosing a partner.



Five Must-Haves

1

TECHNOLOGY AND BUSINESS ACUMEN

IT executives are chartered with meeting business needs by applying technology solutions. So when it's time to choose a development partner, it's critical to find one who can deliver the products you need at the level of quality you would expect from your own staff, at a level of cost you can afford. There is also some onus on the hiring firm to make sure the goals – both business and technology – are sharply defined. Also, be mindful that success can depend on those goals not changing significantly during the development timeframe. Working with a partner that understands your business and how technology is applied in your industry also offers a comfort level that is important while building trust – something that can only happen over time.

2

PASSION FOR PERFECTION

A development partner should demonstrate thought leadership and a passion for delivering not just great code but satisfied customers who perceive them as valued partners. For example, even in initial discussions, a prospective partner should bring relevant technical and domain experts to both demonstrate they can meet client needs and to ensure everyone is on the same page from the very beginning of the sales process.

Passion for perfection engenders a customer-first attitude, demonstrated by doing whatever it takes to get the job done. What are some best practices to look for in a partner?

“Nobody wants to be ‘sold,’ as directly as possible, we want to experience actual software that demonstrates expertise in our desired tech stack. Trust is built on evidence.”

—David Pearah
CEO of the HDF Group

- Proactive communications is job number one. A partner who reduces overhead on your part with frequent – perhaps daily – updates is more likely to stay on track and deliver quality code on time and within budget.
- Education from experience. A development partner with a broad range of skills that can tell a client what works well in a particular instance, what doesn't, and why can evolve from a one-time vendor to an invaluable resource who can guide a business struggling with rapid change.

3

MOBILE-FIRST DESIGN PHILOSOPHY

Since much of the demand for new applications is being driven by the increased use of mobile devices of all kinds, businesses seeking a development partner must ensure that their application code – whether designed for internal, partner or external use – supports the broadest range of mobile devices. That support should start with iOS, Android, Mac and Windows and extend to whatever platforms emerging technologies such as IoT will foist on both IT and user communities.

Mobility's place in business is also being driven by the increasing number of millennials in the workplace. PWC projects that millennials will comprise over 50% of the workforce by 2020, accompanied by a reliance on mobile devices that is often incomprehensible to older co-workers.²

“Bridging the gap between client and service provider can be almost like the Stockholm Syndrome, where the provider becomes empathetic to the client's needs. The best partners provide realistic solutions, yet from angles that the client may not have seen before.”

—Mike Anzola
VP – Mobile Product Management
Citibank



Five Must-Haves

4 TRUST

The issue of trusting a new development partner presents a paradox. Every executive interviewed for this paper cited trust as one of the most important factors while choosing a partner. However, they all have chosen new partners at some time. So how can a partner engender that belief that they are reliable, truthful and able before the first project is completed?

From the sales process through development and billing, development partners should build a relationship as if they are an extension of an organization's own IT staff. If they lack a particular skill or technology required to complete a project, they should make that clear and have a remediation plan in place.

“Crucially important is to maintain short turn-around times and honest dialog. We had one partner who spent a month or so going down blind alleys in the wrong direction.”

**—Mike Sandler
Senior technology executive &
veteran of numerous start-ups**

A common viewpoint can help project an atmosphere of trust.

“Even if a partner has talented people, if they don’t share your mindset they won’t succeed,” says Mark Friedgan, President, Eligo Energy. “It’s important to share how you think about solving problems, building software...things like quality, schedule, responsibility, communications. Everybody has to be on the same page or you get unnecessary conflict which can kill the relationship,” he added.

Of course, nothing succeeds like success – and trust is the byproduct of a series of successful projects. How many successes it takes to develop that trust depends on the relationship more than the software itself.

5 DESIGN THINKING APPROACH

Design thinking isn't new – firms like Apple and design powerhouse IDEO have shown what impact it can have on business outcomes, and global organizations like Toyota, IBM and 3M have been lauded for applying design methods for a broad range of products and services. Simply put, design thinking puts people first, using visualization and other skills to reduce the effort involved to interact with the software – or the device it runs on.

By partnering with a software design firm that adopts this philosophy, a business can gain a myriad of benefits without having to bring a number of new skills in-house. As user interface (UI) and the user experience (UX) increase in importance, styling and visibility will continue to be critical to application success. One critical factor to consider in design is consistency.

“Lack of consistency in a product or suite of products, even in color palette, can make the transition from one function to another difficult and frustrating for even the most seasoned users.”

**—Dan Meier
Sphere Software, Product Designer**

We have discussed the “must-haves,” now we turn to the “must-have-nots.” Here are three issues that could tank a project even before it begins.

1 LACK OF INDUSTRY KNOWLEDGE

There seems to be an infinite number of software development organizations scattered around the globe. While many may have the domain expertise necessary to hit the ground running, some development firms will exaggerate their knowledge of a particular industry if it helps them to land the business, in the hope of acquiring that skill or knowledge once the contract has been signed. Organizations should have the ability to meet the team and gauge their level of knowledge of the topic or industry at hand to establish a comfort level with the potential partner.

2 SHOEHORNING YOUR PROBLEM INTO THEIR SOLUTION

When all you have is a hammer, everything looks like a nail. But if your team isn't ready for an agile development environment or DevOps adoption, that doesn't mean a potential software development partner should push for their client to change their methodologies to suit the outsourcer's needs. Applications written in legacy languages are no less functional, and just because a potential partner is Ruby-heavy does not mean it's the right language for every (or any) of a given business' project plan.

“The difference between a good development partner and a great development partner is a 10 times increase in productivity.”

**—Mike Sandler
Senior technology executive &
veteran of numerous start-ups**

Another red flag is unwillingness of a potential partner to create an up-front demo or evaluation that showcases not only their skills but their understanding of your needs and requirements. This can go a long way towards building trust early in the relationship.

3 LACK OF LOCAL ON-SHORE PRESENCE

Great programmers live everywhere from the US to the Ukraine, and from Ireland to India. Thanks to the near-ubiquity of broadband, it is normal to have a multinational or overseas-based development team working on projects. However, time zones and language barriers present real challenges in terms of miscommunication and delay due to time-shifting. The best way to bypass this problem is to choose a partner with a local presence, with native or bilingual proficiency, and who manages on- and off-shore resources so it is as if everyone is working the same hours.



ABOUT SPHERE SOFTWARE

With over a decade of proven success delivering innovative custom software solutions, Chicago based Sphere builds solutions in software, web, mobile and big data analytics while ensuring the highest level of customer service and satisfaction. Sphere provides unparalleled project management and thought leadership, forming a true partnership with each and every client. Sphere's deep technical acumen includes Ruby on Rails, Python, Scala, Go, Polymer, Clojure, .NET, Java, Node.js to name just a few, and a broad range of industry experience as well.

Sphere believes in complete transparency and open collaboration and communication including open client access to the same internal collaboration tools used by our development team, and Sphere provides full client access to project status updates and source code. To find out how Sphere can help with your software development projects, contact us on our website at www.sphereinc.com



¹ <http://www.businessinsider.com/how-the-internet-of-things-market-will-grow-2014-10>

² <https://www.pwc.com/gx/en/managing-tomorrows-people/future-of-work/assets/reshaping-the-workplace.pdf>